

BASE NUMBER SYSTEM

Number Bases

Different Number Bases are used by the computer, which are

- Base Two – binary
- Base Eight – octal
- Base Sixteen – hexadecimal
- Base Ten -decimal

Conversions

Let's learn other conversions among these four number systems, specifically:

- Binary to Decimal
- Octal to Decimal
- Hexadecimal to Decimal
- Binary to Octal
- Octal to Binary
- Binary to Hexadecimal
- Hexadecimal to Binary
- Octal to Hexadecimal
- Hexadecimal to Octal

Binary to Decimal

- Each binary digit in a binary number has a place value.
- In the number 111, base 2, the digit farthest to the right is in the “ones” place, like the base ten system, and is worth 1.
- Technically this is the 2^0 place.
- The 2nd digit from the right, 111, is in the “twos” place, which could be called the “base” place, and is worth 2.
- Technically this is the 2^1 place.
- In base ten, this would be the “tens” place and would be worth 10.

Binary to decimal

- The 3rd digit from the right, 111, is in the “fours” place, or the “base squared” place, and is worth 4.
- Technically this is the 2^2 place.
- In base ten, this would be the “hundreds” place and would be worth 100.
- The total value of this binary number, 111, is $4+2+1$, or 7.
- In base ten, 111 would be worth $100 + 10 + 1$, or one-hundred eleven.

Binary to Decimal --- solve the problems

- Can you find the decimal values for these given binary values?

➤ **11**

➤ **101**

➤ **110**

➤ **1111**

➤ **11011**

Solutions

- Here are the answers:

- **11 is 3 in base ten**

- **101 is 5**

- **110 is 6**

- **1111 is 15**

- **11011 is 27**

Octal to Decimal

- Octal digits have place values based on the value 8.
- In the number 111, base 8, the digit farthest to the right is in the “ones” place and is worth 1.
- Technically this is the 8^0 place.
- The 2nd digit from the right, 111, is in the “eights” place, the “base” place, and is worth 8.
- Technically this is the 8^1 place.
- The 3rd digit from the right, 111, is in the “sixty-fours” place, the “base squared” place, and is worth 64.
- Technically this is the 8^2 place.
- The total value of this octal number, 111, is $64+8+1$, or seventy-three.

Problems – Octal to decimal

- Can you figure the value for these octal values?

➤ **21**

➤ **156**

➤ **270**

➤ **1164**

➤ **2105**

Here are the answers:

- Here are the answers:

- **21 is 17 in base 10**

- **156 is 110**

- **270 is 184**

- **1164 is 628**

- **2105 is 1093**

Hexadecimal to Decimal

- Hexadecimal digits have place values base on the value 16.
- In the number 111, base 16, the digit farthest to the right is in the “ones” place and is worth 1.
- Technically this is the 16^0 place.
- The 2nd digit from the right, 111, is in the “sixteens” place, the “base” place, and is worth 16.
- Technically this is the 16^1 place.
- The 3rd digit from the right, 111, is in the “two hundred fifty-six” place, the “base squared” place, and is worth 256.
- Technically this is the 16^2 place.
- The total value of this hexadecimal number, 111, is $256+16+1$, or two hundred seventy-three.

Hexadecimal to decimal

- Can you figure the value for these hexadecimal values?

➤ **2A**

➤ **15F**

➤ **A7C**

➤ **11BE**

➤ **A10D**

Solutions

- Here are the answers:

- **2A is 42 in base 10**

- **15F is 351**

- **A7C is 2684**

- **11BE is 4542**

- **A10D is 41229**

Binary to Octal

- The conversion between binary and octal is quite simple.
- Since 2 to the power of 3 equals 8, it takes 3 base 2 digits to combine to make a base 8 digit.

Binary to Octal

➤ 000 base 2 equals 0 base 8

➤ $001_2 = 1_8$

➤ $010_2 = 2_8$

➤ $011_2 = 3_8$

➤ $100_2 = 4_8$

➤ $101_2 = 5_8$

➤ $110_2 = 6_8$

➤ $111_2 = 7_8$

Binary to Octal

- What if you have more than three binary digits, like 110011?
 - Just separate the digits into groups of three from the right, then convert each group into the corresponding base 8 digit.
- 110 011 base 2 = 63 base 8

Binary to Octal

Try these:

➤ 111100

➤ 100101

➤ 111001

➤ 1100101

➤ Hint: when the leftmost group has fewer than three digits, fill with zeroes from the left:

➤ 1100101 = 1 100 101 = 001 100 101

➤ 110011101

Binary to Octal

The answers are:

➤ $111100_2 = 74_8$

➤ $100101_2 = 45_8$

➤ $111001_2 = 71_8$

➤ $1100101_2 = 145_8$

➤ $110011101_2 = 635_8$

Binary to Hexadecimal

- The conversion between binary and hexadecimal is equally simple.
- Since 2 to the power of 4 equals 16, it takes 4 base 2 digits to combine to make a base 16 digit.

Binary to Hexadecimal

➤ 0000 base 2 equals 0 base 16

➤ $0001_2 = 1_{16}$

➤ $0010_2 = 2_{16}$

➤ $0011_2 = 3_{16}$

➤ $0100_2 = 4_{16}$

➤ $0101_2 = 5_{16}$

➤ $0110_2 = 6_{16}$

➤ $0111_2 = 7_{16}$

Binary to Hexadecimal

➤ $1000_2 = 8_{16}$

➤ $1001_2 = 9_{16}$

➤ $1010_2 = A_{16}$

➤ $1011_2 = B_{16}$

➤ $1100_2 = C_{16}$

➤ $1101_2 = D_{16}$

➤ $1110_2 = E_{16}$

➤ $1111_2 = F_{16}$

Binary to Hexadecimal

- If you have more than four binary digits, like 11010111, again separate the digits into groups of four from the right, then convert each group into the corresponding base 16 digit.
 - 1101 0111 base 2 = D7 base 16

Binary to Hexadecimal

Try these:

➤ 11011100

➤ 10110101

➤ 10011001

➤ 110110101

➤ Hint: when the leftmost group has fewer than four digits, fill with zeroes on the left:

➤ 110110101 = 1 1011 0101 = 0001 1011 0101

➤ 1101001011101

Binary to Hexadecimal

The answers are:

➤ $11011100_2 = DC_{16}$

➤ $10110101_2 = B5_{16}$

➤ $10011001_2 = 99_{16}$

➤ $110110101_2 = 1B5_{16}$

➤ $1\ 1010\ 0101\ 1101_2 = 1A5D_{16}$

Octal to Binary

- Converting from Octal to Binary is just the inverse of Binary to Octal.
- For each octal digit, translate it into the equivalent three-digit binary group.
- For example, 45 base 8 equals 100101 base 2

Hexadecimal to Binary

- Converting from Hexadecimal to Binary is the inverse of Binary to Hexadecimal.
- For each “hex” digit, translate it into the equivalent four-digit binary group.
- For example, 45 base 16 equals 01000101 base 2

Octal and Hexadecimal to Binary Exercises

Convert each of these to binary:

➤ 63_8

➤ 123_{16}

➤ 75_8

➤ $A2D_{16}$

➤ 21_8

➤ $3FF_{16}$

Octal and Hexadecimal to Binary Exercises

The answers are:

➤ $63_8 = 110011_2$

➤ $123_{16} = 100100011_2$ (drop leading 0s)

➤ $75_8 = 111101_2$

➤ $A2D_{16} = 110000101101_2$

➤ $21_8 = 10001_2$

➤ $3FF_{16} = 111111111_2$

Hexadecimal to Octal

- Converting from Hexadecimal to Octal is a two-part process.
- First convert from “hex” to binary, then regroup the bits from groups of four into groups of three.
- Then convert to an octal number.

Hexadecimal to Octal

For example:

$$\begin{aligned} &4A3_{16} \\ &= 0100\ 1010\ 0011_2 \\ &= 010\ 010\ 100\ 011_2 \\ &= 2243_8 \end{aligned}$$

Octal to Hexadecimal

- Converting from Octal to Hexadecimal is a similar two-part process.
- First convert from octal to binary, then regroup the bits from groups of three into groups of four.
- Then convert to an hex number.

Octal to Hexadecimal

For example:

$$\begin{aligned} &371_8 \\ &= 011\ 111\ 001_2 \\ &= 1111\ 1001_2 \\ &= F9_{16} \end{aligned}$$

Octal/Hexadecimal Practice

Convert each of these:

➤ $63_8 = \underline{\hspace{2cm}}_{16}$

➤ $123_{16} = \underline{\hspace{2cm}}_8$

➤ $75_8 = \underline{\hspace{2cm}}_{16}$

➤ $A2D_{16} = \underline{\hspace{2cm}}_8$

➤ $21_8 = \underline{\hspace{2cm}}_{16}$

➤ $3FF_{16} = \underline{\hspace{2cm}}_8$

Octal/Hexadecimal Practice

The answers are

$$\blacktriangleright 63_8 = 33_{16}$$

$$\blacktriangleright 123_{16} = 443_8$$

$$\blacktriangleright 75_8 = 3D_{16}$$

$$\blacktriangleright A2D_{16} = 5055_8$$

$$\blacktriangleright 21_8 = 11_{16}$$

$$\blacktriangleright 3FF_{16} = 1777_8$$

Number Base Conversion Summary

- Now you know twelve different number base conversions among the four different bases (2,8,10, and 16)
- With practice you will be able to do these quickly and accurately, to the point of doing many of them in your head!

Decimal numbers

- In the decimal number systems each of the ten digits, 0 through 9, represents a certain quantity. The position of each digit in a decimal number indicates the magnitude of the quantity represented and can be assigned a weight. The weights for whole numbers are positive powers of ten that increases from right to left, beginning with $10^0 = 1$
- 10^5 10^4 10^3 10^2 10^1 10^0
- For fractional numbers, the weights are negative powers of ten that decrease from left to right beginning with 10^{-1} .
- 10^2 10^1 10^0 . 10^{-1} 10^{-2} 10^{-3}
- The value of a decimal number is the sum of digits after each digit has been multiplied by its weights as in following examples.

Problems

1. Express the decimal number 87 as a sum of the values of each digit.

Solution: the digit 8 has a weight of 10, which is 10^1 , as indicated by its position. The digit 7 has a weight of 1, which is 10^0 , as indicated by its position.

$$87 = (8 \times 10^1) + (7 \times 10^0) = (8 \times 10) + (7 \times 1) = 87$$

2. Determine the value of each digit in 939

$$939 = (9 \times 10^2) + (3 \times 10^1) + (9 \times 10^0)$$

Problem using decimal notation

3. Express the decimal number 725.45 as a sum of the values of each digit.

$$\begin{aligned} 725.45 &= (7 \times 10^2) + (2 \times 10^1) + (5 \times 10^0) + (4 \times 10^{-1}) + (5 \times 10^{-2}) \\ &= 700 + 20 + 5 + 0.4 + 0.05 \end{aligned}$$

BINARY NUMBERS

- The binary system is less complicated than the decimal system because it has only two digits, it is a base-two system.
- The two binary digits (bits) are 1 and 0.
- The position of a 1 or 0 in a binary number indicates its weight, or value within the number, just as the position of a decimal digit determines the value of that digit. The weights in a binary number are based on power of two as:
 - 2^4 2^3 2^2 2^1 2^0 . 2^{-1} 2^{-2}
- With 4 digits position we can count from zero to 15. In general, with n bits we can count up to a number equal to $2^n - 1$.
- Largest decimal number = $2^n - 1$

Contd..

- A binary number is a weighted number.
- The right-most bit is the least significant bit (LSB) in a binary whole number and has a weight of $2^0 = 1$.
- The weights increases from right to left by a power of two for each bit.
- The left-most bit is the most significant bit (MSB); its weight depends on the size of the binary number.

Decimal number

Binary number

0

0 0 0 0

1

0 0 0 1

2

0 0 1 0

3

0 0 1 1

4

0 1 0 0

5

0 1 0 1

6

0 1 1 0

7

0 1 1 1

8

1 0 0 0

9

1 0 0 1

10

1 0 1 0

11

1 0 1 1

12

1 1 0 0

13

1 1 0 1

14

1 1 1 0

15

1 1 1 1

Binary-to-Decimal Conversion

- The decimal value of any binary number can be found by adding the weights of all bits that are 1 and discarding the weights of all bits that are 0.
- Example
- Let's convert the binary whole number 101101 to decimal.
- Weight: 2^5 2^4 2^3 2^2 2^1 2^0
- Binary no: 1 0 1 1 0 1
- $101101 = 2^5 + 2^3 + 2^2 + 2^0 = 32 + 8 + 4 + 1 = 45$

Decimal-to-Binary Conversion

- One way to find the binary number that is equivalent to a given decimal number is to determine the set of binary weights whose sum is equal to the decimal number. For example decimal number 9, can be expressed as the sum of binary weights as follows:
 - $9 = 8 + 1$ or $9 = 2^3 + 2^0$
 - Placing 1s in the appropriate weight positions, 2^3 and 2^0 , and 0s in the 2^2 and 2^1 positions determines the binary number for decimal 9.
 - | 2^3 | 2^2 | 2^1 | 2^0 | |
|-------|-------|-------|-------|------------------------|
| 1 | 0 | 0 | 1 | Binary number for nine |

Hexadecimal numbers

- The hexadecimal number system has sixteen digits and is used primarily as a compact way of displaying or writing binary numbers because it is very easy to convert between binary and hexadecimal.
- Long binary numbers are difficult to read and write because it is easy to drop or transpose a bit.
- Hexadecimal is widely used in computer and microprocessor applications. The hexadecimal system has a base of sixteen; it is composed of 16 digits and alphabetic characters.
- The maximum 3-digits hexadecimal number is FFF or decimal 4095 and maximum 4-digit hexadecimal number is FFFF or decimal 65.535

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Binary-to-Hexadecimal Conversion

Simply break the binary number into 4-bit groups, starting at the right-most bit and replace each 4-bit group with the equivalent hexadecimal symbol as in the following example.

Convert the binary number to hexadecimal:

- 1100101001010111

Solution:

1100	1010	0101	0111	
↓	↓	↓	↓	
C	A	5	7	= CA57

Hexadecimal-to-Decimal Conversion

- One way to find the decimal equivalent of a hexadecimal number
 - is to first convert the hexadecimal number to binary
 - and then convert from binary to decimal.
-
- Convert the hexadecimal number 1C to decimal:

$$\begin{array}{cc} 1 & C \\ 0001 & 1100 \end{array} = 2^4 + 2^3 + \mathbf{2^2} = 16 + 8 + 4 = 28$$

Decimal-to-Hexadecimal Conversion

- Repeated division of a decimal number by 16 will produce the equivalent hexadecimal number, formed by the remainders of the divisions.
- The first remainder produced is the least significant digit (LSD).
- Each successive division by 16 yields a remainder that becomes a digit in the equivalent hexadecimal number.
- When a quotient has a fractional part, the fractional part is multiplied by the divisor to get the remainder.

- Convert the decimal number 650 to hexadecimal by repeated division by 16.

$$650 = 40.625 \quad 0.625 \times 16 = 10 = A \text{ (LSD)}$$

$$\begin{array}{r} \hline 16 \end{array} \quad \longrightarrow$$

$$40 = 2.5 \quad 0.5 \times 16 = 8 = 8$$

$$\begin{array}{r} \hline 16 \end{array} \quad \longrightarrow$$

$$2 = 0.125 \quad 0.125 \times 16 = 2 = 2 \text{ (MSD)}$$

$$\begin{array}{r} \hline 16 \end{array} \quad \longrightarrow$$

- The hexadecimal number is 28A

Octal Numbers

- Like the hexadecimal system, the octal system provides a convenient way to express binary numbers and codes.
- However, it is used less frequently than hexadecimal in conjunction with computers and microprocessors to express binary quantities for input and output purposes.
- The octal system is composed of eight digits, which are:
 - 0, 1, 2, 3, 4, 5, 6, 7
- To count above 7, begin another column and start over:
 - 10, 11, 12, 13, 14, 15, 16, 17, 20, 21 and so on.
- Counting in octal is similar to counting in decimal, except that the digits 8 and 9 are not used.

Octal-to-Decimal Conversion

- Since the octal number system has a base of eight, each successive digit position is an increasing power of eight, beginning in the right-most column with 8^0 . The evaluation
- Of an octal number in terms of its decimal equivalent is accomplished by multiplying each digit by its weight and summing the products.
- Let's convert octal number 2374 in decimal number.

Weight 8^3 8^2 8^1 8^0

Octal number 2 3 7 4

$$2374 = (2 \times 8^3) + (3 \times 8^2) + (7 \times 8^1) + (4 \times 8^0) = 1276$$

Decimal-to-Octal Conversion

- A method of converting a decimal number to an octal number is the repeated division-by-8 method, which is similar to the method used in the conversion of decimal numbers to binary or to hexadecimal.
- Let's convert the decimal number 359 to octal. Each successive division by 8 yields a remainder that becomes a digit in the equivalent octal number. The first remainder generated is the least significant digit (LSD).

$$\begin{array}{r} 359 = 44.875 \\ \hline 8 \end{array} \qquad 0.875 \times 8 = 7 \text{ (LSD)} \longrightarrow$$

$$\begin{array}{r} 44 \\ \hline 8 \end{array} = 5.5 \qquad 0.5 \times 8 = 4 \longrightarrow$$

$$\begin{array}{r} 5 \\ \hline 8 \end{array} = 0.625 \qquad 0.625 \times 8 = 5 \text{ (MSD)} \longrightarrow$$

The number is 547.

Octal-to-Binary Conversion

- Because each octal digit can be represented by a 3-bit binary number, it is very easy to convert from octal to binary..

Octal/Binary Conversion

Octal Digit	0	1	2	3	4	5	6	7
Binary	000	001	010	011	100	101	110	111

Let's convert the octal numbers 25 and 140.

2	5		1	4	0
010	101		001	100	000

Binary-to-Octal Conversion

- Conversion of a binary number to an octal number is the reverse of the octal-to-binary conversion.

Let's convert the following binary numbers to octal:

1 1 0 1 0 1

6 5 = 65

1 0 1 1 1 1 0 0 1

5 7 1 = 571